

Route study

This is a revised route study, on request of Christopher Coad, Director of Business Services, who asked that the route study use increased ridership counts for public and non-public routes. Specifically, the impact on transportation is examined for:

- Future ridership for high and middle school buildings
- Future ridership for grades Kindergarten through 5th grade
- Future ridership for pre-school and non-public schools
- Public and non-public school buildings bell schedules
- Alternative routing options
- Estimated cost of each routing option

Determining Ridership

In any routing study, the number of students that ride the buses must be identified first. BBHCSD staff has estimated that 93% of average daily membership (ADM) in grades Kindergarten through grade 5 will be riding the bus equalling 1,514 riders.

Ridership numbers for Middle School students are estimated to be 825 or 92% of the ADM. High school ridership numbers is estimated to be 807 or 60% of its ADM.

BBHCSD has estimated the number of non-public schools' riders to be 298 riders.

BBHCSD has estimated more than 30 pre-school riders, who will be transported on dedicated buses.

Optimal routing occurs when this district can schedule similarly sized groups of students at intervals that allow buses to run multiple routes. In practice, there will always be a certain tier that has more students and requires more buses. Sometimes this occurs because of non-public, special education or vocational routing needs. Sometimes it is purely a result of the number of students that are grouped at a single time. As part of the iterative planning process, the planner focuses on that peak tier and works to either reduce the ridership count or plan extra routing capacity for that tier. The two best ways to make those adjustments are changes in bell times or changes in the number of buses used.

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Routing options and tier arrangement

The following options are a best-case scenario regarding the number of buses used to transport students on each tier and the cost for that option. There may be factors beyond the control of the transportation department, such as road closures, student loading management or administrative student assignments that will require the use of more buses. Fuel costs are rising and expected to be 25% higher for FY2022.

Plans for routing options

Plan D

Uses 48 buses, in a modified two-tier scenario with the elementary students transported on the first tier, middle and high school students on the second tier as follows:

Morning and afternoon transportation

- 34 buses transport middle and high school students, six buses will transport students with special needs to school.
- 34 buses transport elementary school students to school and six buses will transport students with special needs to school.
- Seven buses transport non-public students.

Plan D-1

Uses 48 buses, in a modified two-tier scenario with the middle and high school students transported on the first tier, elementary students on the second tier as follows:

Morning and afternoon transportation

- 34 buses transport middle and high school students, six buses will transport students with special needs to school.
- 34 buses transport elementary school students to school and six buses will transport students with special needs to school.
- Seven buses transport non-public students.
- One bus will transport students to Cuyahoga Valley Career Center (CVCC)

Plan E

Uses a three-tier scenario with 39 buses and the elementary is transported on the first tier, high school on the second tier and middle school students on the third tier. The routing plan is as follows:

Morning and afternoon transportation

- 33 buses transport elementary students on the first tier. To fully utilize bus capacity, an average of 51 students will be assigned to each bus.

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- 23 buses transport high school students on the second tier while 7 buses transport non-public.
- One bus will transport students to CVCC
- 26 buses will transport the middle school students on the third tier.
- Six buses transport students with special needs throughout the tiers including the pre-school students.

Plan E-1

In this plan, we continue the option of a three-tier scenario using 39 buses. The high school students are transported on the first tier and middle school students on the second tier and the elementary students are transported on the last tier. The routing plan is as follows:

Morning and afternoon transportation

- 25 buses transport high school students on the first tier, with an average of 37 students assigned to each bus.
- 23 buses transport middle school students on the second tier while 7 buses transport non-public.
- One bus will transport students to CVCC.
- 33 Buses will be used to transport the elementary students.
- Six buses transport students with special needs throughout the tiers including the pre-school students.

Fleet charts

To illustrate the impact of the routing plans, we have included a fleet assignment chart for each plan that list buses and assignments. These charts are based upon the routing models completed to determine the validity of our assumptions on time and potential for the buses to complete their assignments.

The logistical options have been reviewed and the most viable plans are presented to the Board of Education for consideration. The remaining logistical options have been included in this report to be used for comparison.

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Cost analysis

The cost impact associated with each routing plan is prepared by using the operating cost of each bus as reported on the FY2020 ODE T2.

- 1) Implementing Plan D, (elementary first tier) a two-tier routing plan using 48 buses, has an estimated first year cost of \$3,947,037. The operating cost will continue to be \$2,978,037 annually.
- 2) Implementing Plan D-1, (elementary last tier) a two-tier routing plan using 48 buses, has an estimated first year cost of \$3,996,957. The operating cost will continue to be \$3,027,957 annually.
- 3) Implementing Plan E-1, (elementary/high/middle tiers) a three-tier routing plan using 39 buses, has an estimated operational cost of \$2,527,915 annually.
- 4) Implementing Plan E-1, (high/middle/elementary tiers) a three-tier routing plan using 39 buses, has an estimated operational cost of \$2,654,060 annually.

The estimated annual operating cost impacts do not reflect the capitalization cost of future bus purchases. Bus purchase plans with a 60-bus fleet, 48-route buses and 12 spares should have a goal to replace 7 buses annually to maintain a 10-year replacement cycle and will have an estimated cost of \$710,000 each year.

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Recommendations

- 1) The bell schedule when the new elementary campus opens should plan the order of campuses to be elementary first, high school second and middle school last for both starting and dismissals times. This arrangement allows for more route time in the afternoon and reduces morning layover time between routes.
- 2) Implement Plan E, a three-tier option, that schedules elementary campus on the first tier, and provides flexibility in bus capacity for a continued increase in ridership of public and non-public schools.
- 3) Continue to use group bus stops for all grade levels to minimize the route time for each tier and minimize bus riding time for students.
- 4) Improve bus parking arrangements that reduce the walking distance for students and prioritize bus egress to increase route time efficiency and students' safety.
- 5) The district should consider parents of bus riders subscribing for bus service to plan efficient routes.
- 6) School bus drivers should be counting students monthly and monitor regular bus attendance thoroughly.
- 7) Students that are identified as inactive bus riders should be changed to non-rider status in the routing software for accurate route sheet information.
- 8) Only active riders should be scheduled for the bus routes.
- 9) The bell schedules of all buildings must be correspondingly adjusted. A change in one building bell schedule without a corresponding time for the remaining buildings will have an impact on the bus route time.

Thank you for the opportunity to provide support for the Brecksville Broadview Heights City School District. We welcome the opportunity to help with issues or tasks related to this report if further assistance is desired.